

HOUSE No. 4166

The Commonwealth of Massachusetts

PRESENTED BY:

Michelle L. Ciccolo

To the Honorable Senate and House of Representatives of the Commonwealth of Massachusetts in General Court assembled:

The undersigned legislators and/or citizens respectfully petition for the adoption of the accompanying bill:

An Act to evaluate transportation resiliency and progress towards meeting carbon emissions reductions benchmarks.

PETITION OF:

NAME:	DISTRICT/ADDRESS:	DATE ADDED:
<i>Michelle L. Ciccolo</i>	<i>15th Middlesex</i>	<i>1/15/2025</i>

HOUSE No. 4166

By Representative Ciccolo of Lexington, a petition (accompanied by bill, House, No. 4166) of Michelle L. Ciccolo relative to climate adaptation and transportation infrastructure. Transportation.

The Commonwealth of Massachusetts

In the One Hundred and Ninety-Fourth General Court
(2025-2026)

An Act to evaluate transportation resiliency and progress towards meeting carbon emissions reductions benchmarks.

Be it enacted by the Senate and House of Representatives in General Court assembled, and by the authority of the same, as follows:

1 SECTION 1. Chapter 6C of the General Laws is hereby amended by adding the
2 following section:-

3 Section 80. (a) For the purposes of this section the following words shall, unless the
4 context requires otherwise, have the following meanings:

5 “Adaptation”, adjustments in natural or human systems in response to actual or expected
6 climatic stimuli and associated impacts, including but not limited to changes in processes,
7 practices and protocols to increase resiliency of built and natural structures, moderate potential
8 damages or benefit from opportunities associated with climate change.

9 “Adaptation cost assessment”, a determination of the life cycle financial cost of
10 infrastructure procured and infrastructure improvements made to implement the adaptation
11 strategies identified in a vulnerability assessment. The adaptation cost assessment shall include:

(i) estimated financial costs associated with procurement, construction, maintenance, retrofits and other related improvements made to said infrastructure; and (ii) the estimated financial cost savings resulting from the abated climate risk associated with said infrastructure procurement and improvement. The adaptation cost assessment shall evaluate said infrastructure from its construction or acquisition to its decommissioning. The adaptation cost assessment may include an estimation of the financial value of the co-benefits resulting from said infrastructure procurements and improvements.

“Infrastructure”, including rolling stock, vehicles, equipment, facilities, culverts, roads, bridges, power supplies and other capital asset types, as determined by the secretary.

“Vulnerability assessment”, an evaluation of degree of susceptibility to, or ability to cope with, adverse effects of climate change, such as climate variability and extremes, including an evaluation of adaptive capacity and subsequent adaptation strategies and options for the asset, system, institution or other subject being assessed.

(b) In addition to any vulnerability assessment required pursuant to subsection (b) of section 10 of chapter 21N, the department and each division and authority under the control of the department shall complete a vulnerability assessment and an adaptation cost assessment pertaining to all assets comprising the metropolitan highway system. The vulnerability assessment shall: (i) identify adaptation solutions to be prioritized based on severity of asset vulnerability, level of anticipated climate risk, safety risk to users, riders and workers if vulnerability is not addressed; and (ii) identify whether the asset is located in or serves an environmental justice community. Adaptation solutions shall: (1) identify any opportunities to work in partnership with abutting landowners and municipalities in the implementation of

adaptation solutions; and (2) include the identification of solutions that are reflected in municipal hazard mitigation plans and additional planning efforts related to the municipal vulnerability preparedness grant program. Every 4 years, the department shall review and update the vulnerability assessment and adaptation cost assessment required pursuant to this subsection.

(c) The department and each division and authority under the control of the department shall complete a vulnerability assessment and an adaptation cost assessment pertaining to all equipment and mass transportation facilities, as defined in section 1 of chapter 161A. Each vulnerability assessment shall: (i) identify adaptation strategies to be prioritized based on severity of asset vulnerability, level of anticipated climate risk and safety risk to users, riders and workers if vulnerability is not addressed; (ii) identify whether the asset is located in or serves an environmental justice community; (iii) identify the authority's plan to provide replacement or alternative service to sustain the current level of service to riders in the event assets are unsuitable to operate due to damage associated with climate risks, including, but not limited to, flooding, extreme heat, excessive snow and ice and other extreme weather events; and (iv) include the cost of any temporary or extended use facilities required to provide ongoing access or detours for roads and bridges during their replacement or repair. Adaptation solutions shall: (1) identify any opportunities to work in partnership with abutting landowners and municipalities; and (2) include identification of solutions that are reflected in municipal hazard mitigation plans and additional planning efforts related to the municipal vulnerability preparedness grant program. 4 years, the department and each division and authority under the control of the department shall review and update the vulnerability assessments and cost adaptation assessments required pursuant to this subsection.

(d) The vulnerability assessments and adaptation cost assessments required pursuant to subsections (b) and (c) shall be completed within 18 months of the enactment of this section. Every 4 years following the enactment of this section, the secretary of the department shall provide a report detailing progress toward the vulnerability and adaptation cost assessments required pursuant to said subsections (b) and (c) to the clerks of the house of representatives and the senate, the chairs of the house and senate committees on ways and means and the chairs of the joint committee on transportation.

SECTION 2. The second paragraph of subsection (g) of section 5 of chapter 161A of the General Laws, as appearing in the 2022 Official Edition, is hereby amended by inserting after the word “act” the following words:- , capital investments identified as priority adaptation strategies in the department’s vulnerability assessment.

SECTION 3. Chapter 161B of the General Laws is hereby amended by adding the following section:-

Section 28. (a) For the purposes of this section the following words shall, unless the context requires otherwise, have the following meanings:

“Adaptation”, adjustments in natural or human systems in response to actual or expected climatic stimuli and associated impacts, including but not limited to changes in processes, practices and protocols to increase resiliency of built and natural structures, moderate potential damages or benefit from opportunities associated with climate change.

“Adaptation cost assessment”, a determination of the life cycle financial cost of infrastructure procured and infrastructure improvements made to implement the adaptation strategies identified in a vulnerability assessment. The adaptation cost assessment shall include:

(i) estimated financial costs associated with procurement, construction, maintenance, retrofits and other related improvements made to said infrastructure; and (ii) the estimated financial cost savings resulting from the abated climate risk associated with said infrastructure procurement and improvement. The adaptation cost assessment shall evaluate said infrastructure from its construction or acquisition to its decommissioning. The adaptation cost assessment may include an estimation of the financial value of the co-benefits resulting from said infrastructure procurements and improvements.

“Infrastructure”, including rolling stock, facilities, vehicles, equipment, culverts, roads, bridges, power supplies and other capital asset types, as determined by the council.

“Vulnerability assessment”, an evaluation of degree of susceptibility to, or ability to cope with, adverse effects of climate change, such as climate variability and extremes, including an evaluation of adaptive capacity and subsequent adaptation strategies and options for the asset, system, institution or other subject being assessed.

(b) All regional transit authorities, as established in section 2, shall complete a vulnerability assessment and an adaptation cost assessment. The vulnerability assessment and adaptation cost assessment shall pertain to all equipment and mass transportation facilities as defined in section 1 of chapter 161A. The vulnerability assessment shall: (i) identify adaptation strategies to be prioritized based on severity of asset vulnerability, level of anticipated climate risk and safety risk to riders and workers if vulnerability is not addressed; (ii) identify whether the asset is located in or serves an environmental justice community; and (iii) identify the authority’s plan to provide replacement or alternative service to sustain the current level of service to riders in the event assets are unsuitable to operate due to damage associated with

climate risks, including but not limited to flooding, extreme heat, excessive snow and ice and other extreme weather events. Adaptation solutions shall: (1) identify any opportunities to work in partnership with abutting landowners and municipalities; and (2) include identification of solutions that are reflected in municipal hazard mitigation plans and additional planning efforts related to the municipal vulnerability preparedness grant program. Every 2 years, each regional transit authority shall review and update the vulnerability assessments and cost adaptation assessments required pursuant to this subsection.

(c) The vulnerability assessment and adaptation cost assessment required pursuant to subsection (b) shall be completed within 18 months of the enactment of this section. Every 4 years following the enactment of this section, the secretary of the regional transit authority council shall provide a report detailing progress toward the vulnerability and adaptation cost assessments required pursuant to said subsection (b) to the clerks of the house of representatives and the senate, the chairs of the house and senate committees on ways and means and the chairs of the joint committee on transportation.

SECTION 4. Chapter 465 of the acts of 1956 is hereby amended by adding the following section:-

Section 37. (a) For the purposes of this section the following words shall, unless the context requires otherwise, have the following meanings:

“Adaptation”, adjustments in natural or human systems in response to actual or expected climatic stimuli and associated impacts, including but not limited to changes in processes, practices and protocols to increase resiliency of built and natural structures, moderate potential damages or benefit from opportunities associated with climate change.

“Adaptation cost assessment”, a determination of the life cycle financial cost of infrastructure procured and infrastructure improvements made to implement the adaptation strategies identified in a vulnerability assessment. The adaptation cost assessment shall include: (i) estimated financial costs associated with procurement, construction, maintenance, retrofits and other related improvements made to said infrastructure; and (ii) the estimated financial cost savings resulting from the abated climate risk associated with said infrastructure procurement and improvement. The adaptation cost assessment shall evaluate said infrastructure from its construction or acquisition to its decommissioning. The adaptation cost assessment may include an estimation of the financial value of the co-benefits resulting from said infrastructure procurements and improvements.

“Infrastructure”, including rolling stock, facilities, vehicles, equipment, culverts, roads, bridges, power supplies and other capital asset types, as determined by the Authority.

“Vulnerability assessment”, an evaluation of degree of susceptibility to, or ability to cope with, adverse effects of climate change, such as climate variability and extremes, including an evaluation of adaptive capacity and subsequent adaptation strategies and options for the asset, system, institution or other subject being assessed.

(b) The Authority shall complete a vulnerability assessment and an adaptation cost assessment pertaining to all equipment and mass transportation facilities, as defined in section 1 of chapter 161A of the General Laws. Each vulnerability assessment shall: (i) identify adaptation strategies to be prioritized based on severity of asset vulnerability, level of anticipated climate risk and safety risk to users, riders and workers if vulnerability is not addressed; (ii) identify whether the asset is located in or serves an environmental justice community; (iii) identify the

144 authority's plan to provide replacement or alternative service to sustain the current level of
145 service to riders in the event assets are unsuitable to operate due to damage associated with
146 climate risks, including, but not limited to, flooding, extreme heat, excessive snow and ice and
147 other extreme weather events; and (iv) include the cost of any temporary or extended use
148 facilities required to provide ongoing access or detours for roads and bridges during their
149 replacement or repair. Adaptation solutions shall: (1) identify any opportunities to work in
150 partnership with abutting landowners and municipalities; and (2) include identification of
151 solutions that are reflected in municipal hazard mitigation plans and additional planning efforts
152 related to the municipal vulnerability preparedness grant program. Every 4 years, the Authority
153 shall review and update the vulnerability assessments and cost adaptation assessments required
154 pursuant to this subsection.

155 (c) The vulnerability assessment and adaptation cost assessment required pursuant to
156 subsection (b) shall be completed within 18 months of the enactment of this subsection. Every 4
157 years following the enactment of this section, the Authority shall provide a report detailing
158 progress toward the vulnerability and adaptation cost assessments required pursuant to said
159 subsection (b) to the clerks of the house of representatives and the senate, the chairs of the house
160 and senate committees on ways and means and the chairs of the joint committee on
161 transportation.

162 SECTION 5. (a) There is hereby established a special commission on transportation
163 finance for "net-zero by 2050" to investigate, study and make recommendations on the
164 development and deployment of comprehensive and regionally equitable public transportation
165 pricing, roadway pricing and congestion pricing to further the climate goals of the
166 commonwealth. The commission shall be composed of the following members: 5 members who

167 shall not be employees of the executive branch and who shall reside in different geographic
168 regions of the commonwealth, 1 of whom shall be a representative of the construction industries
169 of the commonwealth, to be appointed by the governor to serve terms of 2 years; 3 members,
170 who shall not be members of the general court and who shall reside in different geographic
171 regions of the commonwealth, to be appointed by the president of the senate to serve terms of 2
172 years; 3 members, who shall not be members of the general court and who shall reside in
173 different geographic regions of the commonwealth, to be appointed by the speaker of the house
174 of representatives to serve terms of 2 years; a representative of the Massachusetts Taxpayers
175 Foundation, Inc.; a representative of the Massachusetts Business Roundtable, Inc.; a
176 representative of the Massachusetts Municipal Association, Inc.; a representative of the regional
177 transit authorities as chosen by the secretary of transportation; a representative of the regional
178 planning agencies as chosen by the Massachusetts Association of Regional Planning Agencies;
179 and a representative of the Massachusetts Bay Transportation Authority advisory board,
180 established in section 7A of chapter 161A of the General Laws. Each of the members of the
181 commission shall be an expert with experience in the fields of law or public policy,
182 transportation planning or the design and construction of transportation projects. One of the
183 members appointed by the governor, 1 of the members appointed by the president of the senate
184 and 1 of the members appointed by the speaker of the house of representatives shall be
185 representatives of the Massachusetts business community. One of the members appointed by the
186 governor, 1 of the members appointed by the president of the senate and 1 of the members
187 appointed by the speaker of the house of representatives shall be representatives of
188 environmental organizations, planning organizations, transportation consumer organizations or
189 other public interest organizations. One of the members appointed by the governor shall be an

expert in the field of management consulting or organizational change. One of the members appointed by the governor shall be an expert in the field of public finance. One of the members shall be appointed by the governor to serve as chairperson of the commission. The members of the commission shall be appointed no later than 2 months following the passage of this act.

(b) The commission shall:

(i) endeavor to understand the current transportation financial landscape, by identifying and providing for current transportation funding forecasts, including but not limited to: (i) providing estimates of annual operation and maintenance costs for roadways; (ii) providing estimates of annual revenue with consideration of declining motor vehicle fuel excise revenue due to vehicle electrification; (iii) providing traffic forecasts, including forecasts of traffic diversion impacts; (iv) providing year to year estimates of annual operation and maintenance costs of the Massachusetts Bay Transit Authority and regional transit authorities; and (v) providing estimates of annual revenue, and the methodology employed for such estimates, including an analysis of revenue from the “fair share” amendment for the repair and maintenance of roads, bridges and public transportation;

(ii) provide specific operating and capital funding options for the transportation sector to align with the commonwealth’s year 2050 requirements for net-zero greenhouse gas emissions and provide for transportation climate resiliency, including identifying and providing for sources of funding for such costs for meeting the 2025, 2030, 2035, 2040, 2045 and 2050 statewide greenhouse gas emissions limits. The commission shall: (i) identify and analyze physical, technological, legal and other issues or requirements related to roadway pricing in the commonwealth; (ii) propose detailed specifications and regionally equitable locations for toll

gantries and other equipment necessary to assess and collect tolls; (iii) review roadway pricing scenarios under the Federal Highway Administration's Value Pricing Pilot Program; (iv) propose equitable increase in transportation network company fees, including increased ride share fees and excise taxes on out-of-state operators; (v) propose any other revenue sources and strategies that may be needed to meet the commonwealth's 2050 emissions mandates; (vi) provide a regional and social equity analysis with specific recommendations related to mitigating adverse impacts for low income residents; (vii) identify all local, state and federal approvals necessary to deploy new tolls and other roadway pricing mechanisms on relevant roadways; and (viii) take into consideration roadway and congestion pricing programs in other jurisdictions throughout the United States and the world;

(iii) identify costs to ensure a reliable, safe and resilient public transportation system that aligns with the commonwealth's year 2050 requirements for net-zero greenhouse gas emissions, including identifying and providing for sources of funding for such costs for meeting the 2025, 2030, 2035, 2040, 2045 and 2050 statewide greenhouse gas emissions limits. The commission shall: (i) study commute and demand patterns for public transit entities; (ii) study economic development and housing patterns and projections and the impact each has on public transit demand; (iii) review the commonwealth's laws regarding emissions reductions within the transportation sector; (iv) determine fare structures for all modes of transit of the Massachusetts Bay Transportation Authority and regional transit authorities that account for commute patterns and demand, economic development and housing patterns and emissions reduction requirements; (v) examine the feasibility of means-tested fares; (vi) provide estimates of costs to implement the adaptation needs identified in the vulnerability assessment; (vii) provide estimates of costs of annual revenue; (viii) provide ridership forecasts; (ix) provide a regional and social equity

analysis with specific recommendations related to mitigating adverse impacts; (x) identify all local, state and federal approvals necessary to deploy new fare structures at regional transit authorities and the Massachusetts Bay Transportation Authority; and (xi) provide potential impacts on vehicular emissions reduction;

(iv) analyze the efficacy and financial sufficiency of the commonwealth's current chapter 90 program in meeting the goal of net zero by 2050 and climate resiliency; provided, that the commission shall include in its review the sufficiency of the chapter 90 program's current funding formula based on local road mileage, population and employment and alternatives thereto, including the additional or alternative factor of a municipalities' equalized value in the formula;

(v) consult and collaborate with all of the commonwealth's regional transportation agencies, regional planning agencies, regional economic development organizations, the Massachusetts Municipal Association, Inc., the commonwealth's gateway cities and the city of Boston to address geographical transportation financing challenges and solutions that vary within and among the commonwealth's regions; and

(vi) consult and collaborate with community-based organizations and communities of color and diversity regarding social equity issues impacting transportation sector financing including, but not limited to, social equity issues for communities underserved by the current transportation financing system and most directly impacted by congestion and greenhouse gas emissions. The commission shall, on an ongoing basis, evaluate its efforts to provide for such consultation and collaboration.

(c) The commission may also: (i) propose any other revenue sources and strategies that may be necessary to meet the commonwealth's 2050 emissions mandates; (ii) provide year-to-year direct measurements and estimates of annual operation and maintenance costs; and (iii) propose methods of distributing methods of raised revenue that are in line with the program's goals.

(d) The commission may investigate, study and make recommendations on additional mobility methods as necessary.

(e) Not later than 12 months following the passage of this act, the commission shall develop an interim initial report with recommendations on Massachusetts Bay Transit Authority and the regional transit authorities financing.

(f) Not later than 18 months following the passage of this act, the commission shall develop a final report detailing its findings, including a draft of any legislation required to implement its recommendations, to the governor, the secretary of transportation, the clerks of the house of representatives and the senate, the house and senate committees on ways and means and the joint committee on transportation.

(g) Not later than three months from the passage of this act, the treasurer shall transfer from the Massachusetts Transportation Trust Fund \$200,000 for the administration of the commission; provided, that the commission shall have an budget of \$100,000 for the duration of the project and shall have access to the resources and staff of the employees of the Massachusetts Department of Transportation and the Massachusetts Bay Transit Authority and may employ such additional staff or consultants as it may deem necessary.

SECTION 6. Section 5 shall take effect 6 months following the passage of this act.